

WHAT MRI DETECTS – THE ¹H PROTON SPECTRUM

MRI detects protons (¹H) in different molecular environments.
These environments cause protons to resonate at slightly different frequencies (chemical shift).

KEY POINT

In clinical MRI we image the dominant water and fat signals. In MR Spectroscopy (MRS) we resolve many smaller metabolite peaks.

WHAT IS DETECTED?

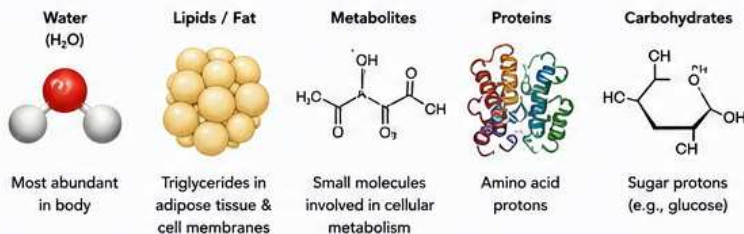
Conventional MRI uses ¹H (hydrogen) protons.

Different molecules and tissues place those protons in unique electronic environments, causing them to precess at slightly different frequencies.

This creates the MR spectrum.



MAIN ¹H PROTON SOURCES IN THE BODY

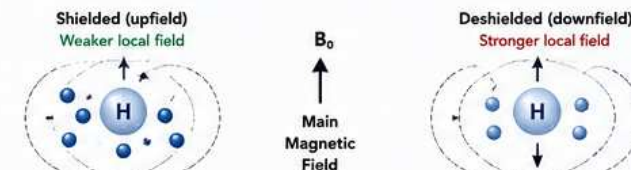


WHY DO PROTONS SHOW UP AT DIFFERENT FREQUENCIES?

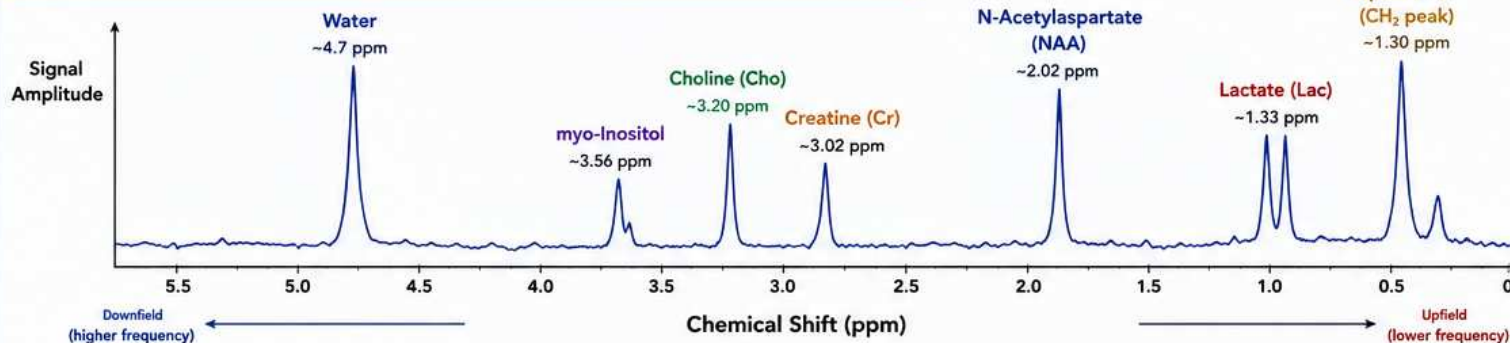
Electrons around nuclei create tiny local magnetic fields.

More shielding → weaker local field → lower frequency (upfield)

Less shielding → stronger local field → higher frequency (downfield)



TYPICAL ¹H MR SPECTROSCOPY (MRS) SPECTRUM (e.g., Brain)



ABOUT CHEMICAL SHIFT (ppm)

Chemical shift (δ) is reported in parts per million (ppm) relative to a reference (usually TMS or water).

- Higher ppm (left) = downfield = higher frequency
- Lower ppm (right) = upfield = lower frequency

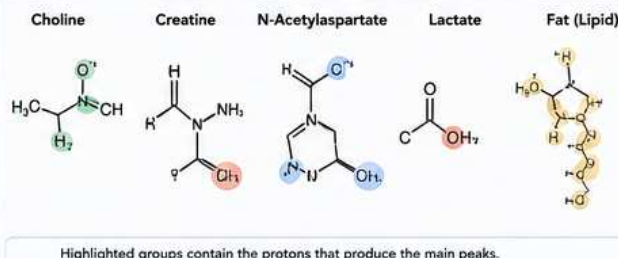
Field strength affects frequency (Hz), but NOT the ppm position.

This spectrum was acquired at ~3 Tesla (typical). Exact ppm values can vary slightly with field strength, pH and temperature.

MAJOR PEAKS: WHAT THEY REPRESENT

Peak (Abbrev.)	Chemical Shift (ppm)	What It Represents	Where Found / Relevance
Water	~4.7	Water protons (H ₂ O)	All body tissues; dominant in MRI signal
myo-Inositol (mi)	~3.56	Alcohol protons in inositol ring	Glial cells; marker in brain disorders
Choline (Cho)	~3.20	Methyl protons in choline compounds	Cell membrane turnover; ↑ in tumors
Creatine (Cr)	~3.02	Methyl protons in creatine	Energy metabolism; relatively stable
N-Acetylaspartate (NAA)	~2.02	Methyl protons in NAA	Neuronal marker; ↓ in neuronal loss
Lactate (Lac)	~1.33	Methyl protons in lactate (doublet)	Anaerobic metabolism; ↑ in ischemia/tumor
Lipids / Fat (CH ₂)	~1.30	Methylene protons in fatty acyl chains	Adipose tissue, membranes, necrosis

EXAMPLE MOLECULAR SOURCES (SIMPLE VIEW)



Highlighted groups contain the protons that produce the main peaks.

WHERE THESE PROTONS COME FROM

Brain	Muscle	Adipose (Fat)	Liver	Tumor / Lesion
Neurons, glia, metabolites, water	Water, creatine, lactate	Triglycerides, lipid protons	Water, lipids, metabolites, glycogen	Altered choline, lactate, lipids, reduced NAA



KEY TAKEAWAY

MRI detects ¹H protons. Because protons in different chemical environments resonate at different frequencies, the scanner can separate those frequencies using the Fourier Transform to produce the spectrum above. This is the foundation of both imaging (water vs. fat) and spectroscopy (metabolite analysis).

From Time Domain (FID)



→ FFT →

To Frequency Domain

